CURBCOUNT · USER REQUIREMENTS

CurbCount user requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/curbcount		
Source	docs/03a-user-requirements.md @ 82b4850		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-31	Amish Chadha	Opened in the kit 1.9.1 upgrade; the needs of the six users in CBC-PRB-001 written from their side, split out of the old requirements

CurbCount user requirements

What the people who would use or live with a CurbCount counter need, written from their side. Product requirements that meet these needs are in CBC-PRS-001 (`03-requirements.md`) and the links are in the traceability matrix (`03b-traceability.md`). Method: STANDARDS section 22. These needs were drawn from the user table in CBC-PRB-001 and have not yet been checked with a co-design partner (CBC-DDR-002, O1 names the first candidate).

Table 1. User requirements.

ID	Category	Title	Need	Priority	Source
UR-001	Deployment and assembly	Fit to an existing pole	Two people can fit the counter to a street pole that is already there, without drilling, welding or touching its wiring	High	Asset owner (PRB users table)
UR-002	Deployment and assembly	Light enough to carry up	The parts can be lifted and fitted from a mobile platform by one person at a time	Medium	Asset owner (PRB users table)
UR-003	Performance	Count by mode and direction	People on foot, people on bicycles and scooters, and motor vehicles are counted separately, for each direction along the street	High	City transport planner; road safety engineer
UR-004	Performance	Counts close to a manual count	The counts are close enough to a manual count to show whether a change to the street made a difference	High	City transport planner; researcher or student
UR-005	Performance	Whole street from one pole	One counter sees the sidewalk, the bike lane and the nearest traffic lane	High	City transport planner
UR-006	Safety	No images, no identity	No image, sound or personal identifier leaves the device, and the sensor cannot show a face or a number plate even if someone wanted it to	High	Residents' group or NGO
UR-007	Usability	Open, documented counts	The counts arrive in a short record whose format is published, in fixed time bins	Medium	Researcher or student; city transport planner
UR-008	Usability	Set up and aim check on site	The crew can check the aim and confirm an uplink on site with a laptop or phone, with no special software	Medium	City transport planner
UR-009	Environment and durability	Counts through the seasons	The counter works in the temperatures of the site through the year, in sun and in shade	High	City transport planner; business district manager
UR-010	Environment and durability	Runs without pole power	The counter runs from its own panel and cell, including through dull winter days and short spells with no sun	High	Asset owner (street lights switch off in the day)
UR-011	Environment and durability	Survives five years outdoors	Rain, dust, sun and wind do not stop it working for years	High	Asset owner
UR-012	Maintenance	Rare, simple service	A cell change is the only routine service in five years	Medium	Asset owner
UR-013	Cost	Cheap enough for many poles	The parts cost is low enough that a city, a group or a university can deploy several counters	High	City transport planner; residents' group; business district manager
UR-014	Regulatory	Legal radio use	The radio link stays within the licence-free limits where it is used	Medium	City transport planner

ID	Category	Title	Need	Priority	Source
UR-015	Safety	Stays on the pole, safe to pass	The counter does not fall, twist or shed parts onto the street in wind, and adds no hazard for a person below	High	Asset owner; residents' group or NGO
UR-016	Usability	Public can see what it is	A resident can read what the device counts and find the open design from the pole	Medium	Residents' group or NGO

Categories: deployment and assembly, performance, usability, safety, environment and durability, logistics, maintenance, cost, regulatory. Priorities: High, Medium, Low.